

Canada-U.S. Trade Shock Observatory

2025 Section 232 core steel - initial analytical release

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Independent Portfolio Project

A reproducible account of U.S. imports for consumption from Canada, built from official Census records and matched to the 2025 tariff schedule.

2025 IMPORT VALUE	ANNUAL CHANGE	2025 CALCULATED DUTY
\$4.51B	-36.8%	\$1.07B

Main result

The value of Canadian core-steel imports in scope was 36.8% lower in 2025 than in 2024. Year-over-year declines were larger in the later tariff periods, although the comparison by itself cannot establish that the tariff caused the change.

Release date: 2026-09-12 | Validation: PASSED | 96 archived API responses

1. Scope and methodology

The observatory measures monthly U.S. imports for consumption of Canadian core steel. Product scope follows the original Section 232 steel ranges and excludes later derivative articles.

Dimension	Initial release specification
Trade flow	U.S. imports for consumption from Canada (Census country code 1220)
Coverage	January 2024 through December 2025
Product universe	746 HTS10 lines mapped to 168 HS6 codes
Outcomes	Import value, dutiable value, calculated duty; quantity where reported
Data grain	HS6-country-month for full scope; HTS10-country-month for pilot
Source handling	Exact JSON responses retained with SHA-256 hashes; API key excluded

Policy timeline

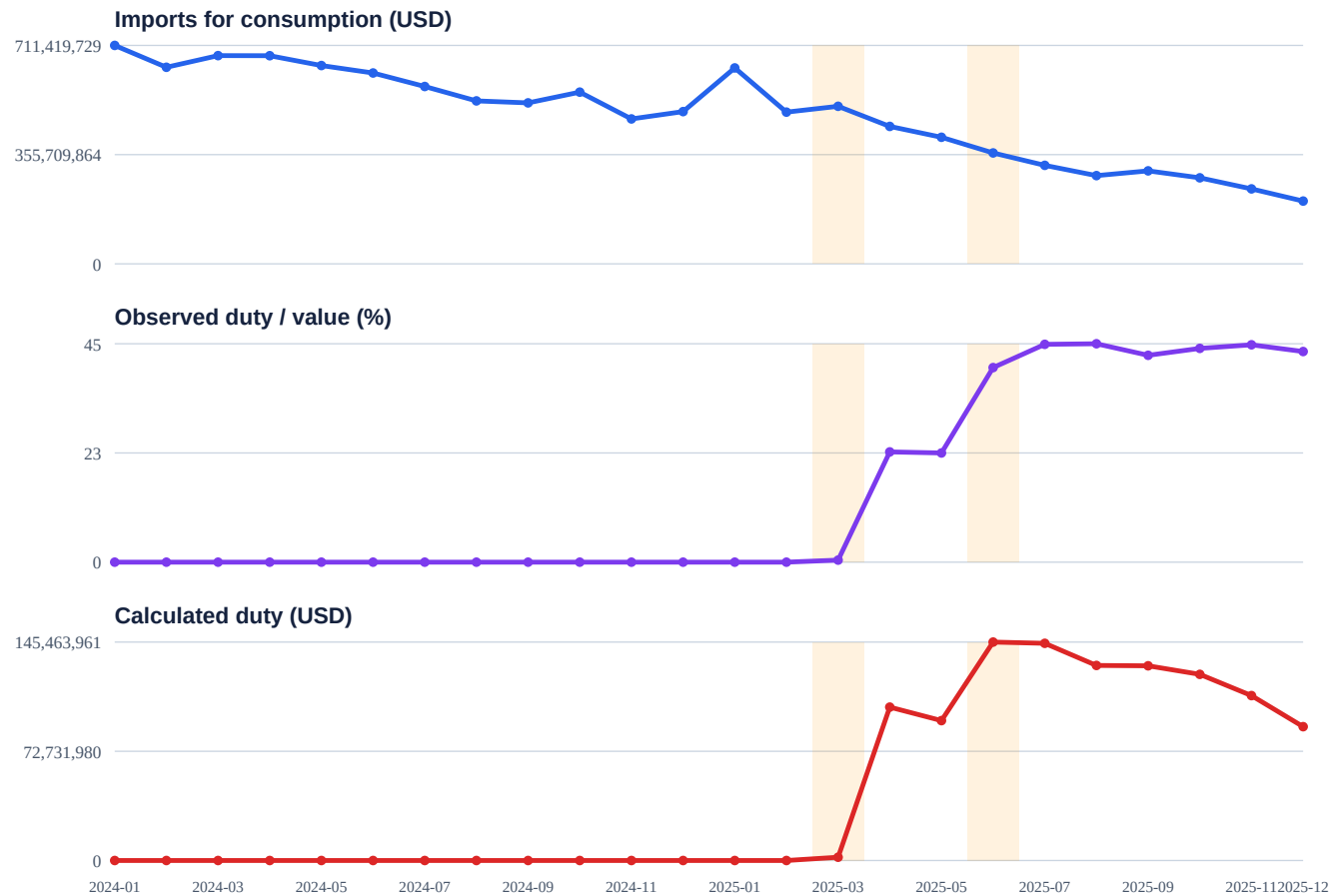
Before Mar. 12, 2025	0%	Canadian exemption in effect
Mar. 12 - Jun. 3, 2025	25%	Exemption terminated
From Jun. 4, 2025	50%	Additional rate increased

March and June are transition months. The panel reports start-of-month, end-of-month, and calendar-day-weighted statutory rates, but deliberately leaves the clean-month treatment rate blank. Day weighting is descriptive and does not assume imports were evenly distributed across days.

Census returns both a rate-provision total row (RP='-') and its detailed breakdown. The production panel uses the total row and validates this behavior to prevent double-counting.

2. Full core-steel results

Import value generally declined during 2025, while reported calculated duty increased after the March and June policy changes. The duty-to-value series divides Census calculated duty by total import value.



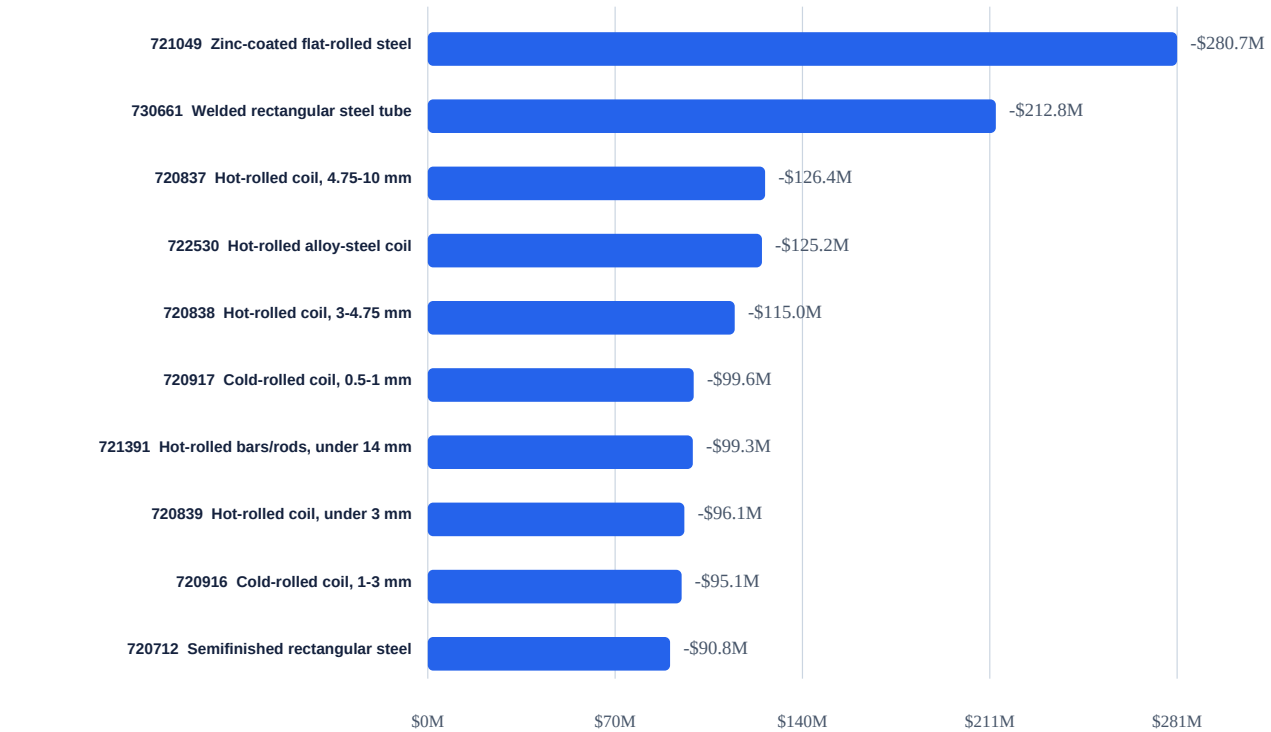
Orange shading marks March and June 2025 transition months.

2025 clean window	Import value	vs. same 2024 months	Duty / value
Pre-tariff: Jan.-Feb.	\$1.132B	-16.2%	0.0%
25%: Apr.-May	\$859.8M	-35.0%	22.7%
50%: Jul.-Dec.	\$1.640B	-48.1%	44.3%

Reading the table: the year-over-year decline is larger in each successive clean policy window. Prices, demand, seasonality, inventory decisions, product substitution, and advance shipments may also matter. A causal estimate would require a comparison group and a specified counterfactual.

3. Products contributing most to the decline

The chart ranks HS6 products by the reduction in annual import value from 2024 to 2025. Together, the top ten account for a substantial share of the aggregate decline, led by zinc-coated flat-rolled steel (HS6 721049) and welded rectangular tubing (HS6 730661).

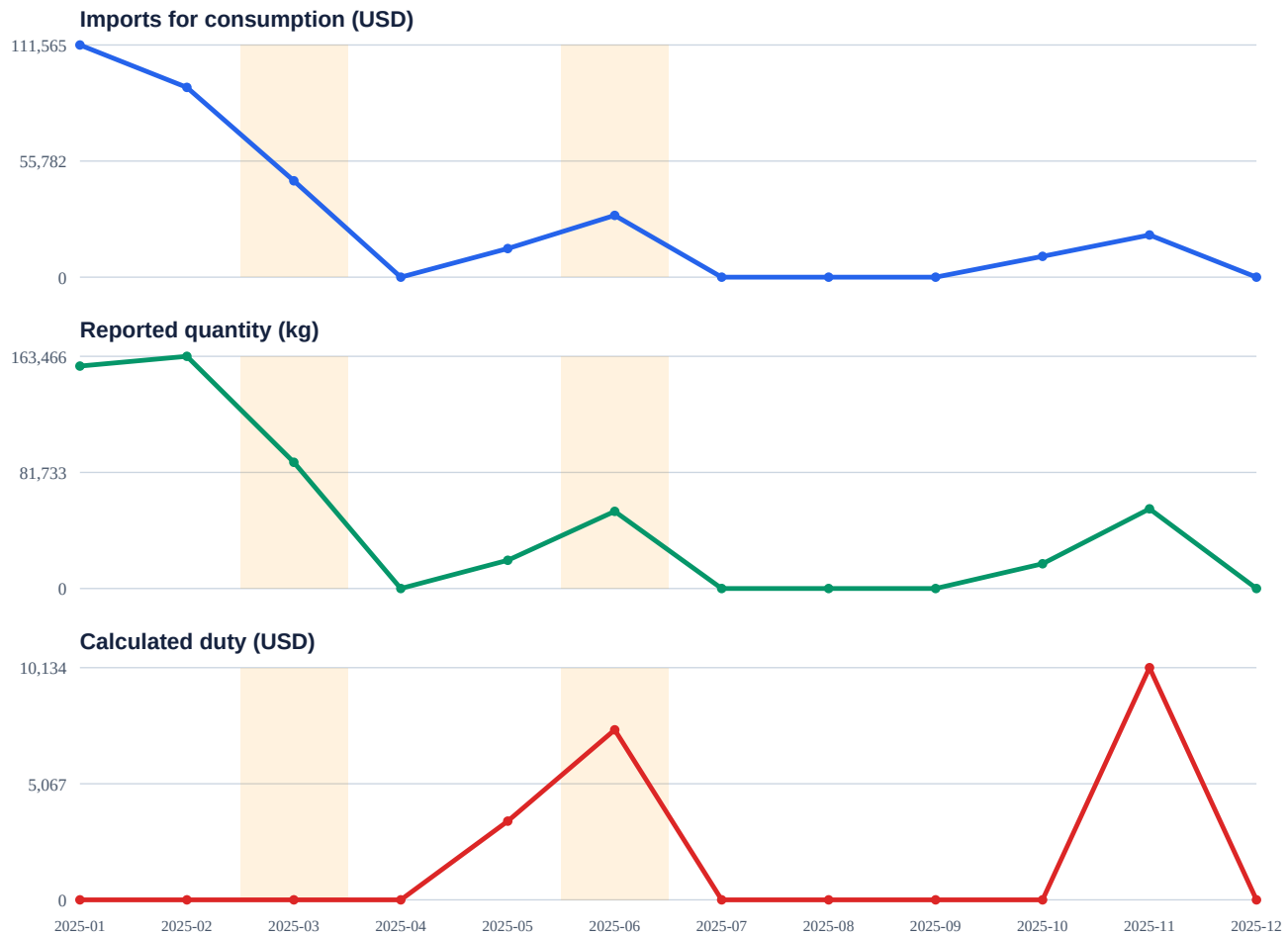


Rank	HS6	Product	2024 value	2025 value	Change	Share
1	721049	Zinc-coated flat-rolled steel	\$817.3M	\$536.6M	-\$280.7M	10.7%
2	730661	Welded rectangular steel tube	\$447.9M	\$235.1M	-\$212.8M	8.1%
3	720837	Hot-rolled coil, 4.75-10 mm	\$241.7M	\$115.3M	-\$126.4M	4.8%
4	722530	Hot-rolled alloy-steel coil	\$233.2M	\$108.0M	-\$125.2M	4.8%
5	720838	Hot-rolled coil, 3-4.75 mm	\$222.9M	\$107.9M	-\$115.0M	4.4%
6	720917	Cold-rolled coil, 0.5-1 mm	\$214.1M	\$114.5M	-\$99.6M	3.8%
7	721391	Hot-rolled bars/rods, under 14 mm	\$300.5M	\$201.2M	-\$99.3M	3.8%
8	720839	Hot-rolled coil, under 3 mm	\$181.4M	\$85.3M	-\$96.1M	3.7%
9	720916	Cold-rolled coil, 1-3 mm	\$219.4M	\$124.3M	-\$95.1M	3.6%
10	720712	Semifinished rectangular steel	\$171.7M	\$80.9M	-\$90.8M	3.5%

This is an accounting decomposition of the observed value change, not a causal estimate of product-level tariff effects.

4. HTS10 pilot: 7208101500

Flat-rolled iron or nonalloy steel coils, 600 mm or more wide, hot-rolled, pickled, with patterns in relief. Unlike the HS6 aggregate release, this line reports quantity in kilograms.



Month	Value	Quantity (kg)	Duty	Policy
2025-01	\$111,565	156,594	\$0	no section232 tariff
2025-02	\$91,190	163,466	\$0	no section232 tariff
2025-03	\$46,308	88,864	\$0	transition 0 to 25
2025-04	\$0	0	\$0	25 percent
2025-05	\$13,746	19,876	\$3,436	25 percent
2025-06	\$29,683	54,317	\$7,421	transition 25 to 50
2025-07	\$0	0	\$0	50 percent
2025-08	\$0	0	\$0	50 percent
2025-09	\$0	0	\$0	50 percent
2025-10	\$10,000	17,423	\$0	50 percent
2025-11	\$20,267	56,096	\$10,134	50 percent
2025-12	\$0	0	\$0	50 percent

5. Quality, limitations, and reproducibility

Release quality controls

Check	Result
Automated tests	18 passed
Official HS6-month rows	3,694
Months	24
Raw Census snapshots	96
Snapshot integrity	SHA-256 verified; API key excluded from metadata
Transition handling	March and June 2025 flagged; excluded from clean comparisons
HS6 quantity	Unavailable in aggregate API rows; stored as null, not zero

Limitations

This release covers one trade flow over two years. It does not measure welfare, tariff pass-through, trade diversion, domestic output, employment, or a causal treatment effect. Product coverage follows the original core-steel ranges at HS6; later derivative additions are outside the initial scope.

Reproduction

Run `python scripts/build_release.py --reuse-snapshots` to rebuild the processed files from the archived source responses. Run `python -m unittest discover -s tests -v` with the project source path configured to repeat the validation tests.

Official sources

- [U.S. Census International Trade API](#)
- [Census API variable definitions](#)
- [January 2025 Census commodity concordance](#)
- [February 2025 steel proclamation](#)
- [June 2025 steel and aluminum proclamation](#)

Publication note

Recommended citation: Roy, Ridoy. 2026. *Canada-U.S. Trade Shock Observatory: 2025 Section 232 Core Steel, Initial Analytical Release*. Data derived from the U.S. Census Bureau and official presidential proclamations.